



University of Frontier Technology, Bangladesh

Department of Internet of Things and Robotics Engineering

Course Code	CC 483
Course Title	Cloud Computing

LAB REPORT-11 Solution

AZ-104: Implementing Data Protection

Azure Backup & Site Recovery — IaaS Virtual Machine Protection

Submitted To

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Date of Submission

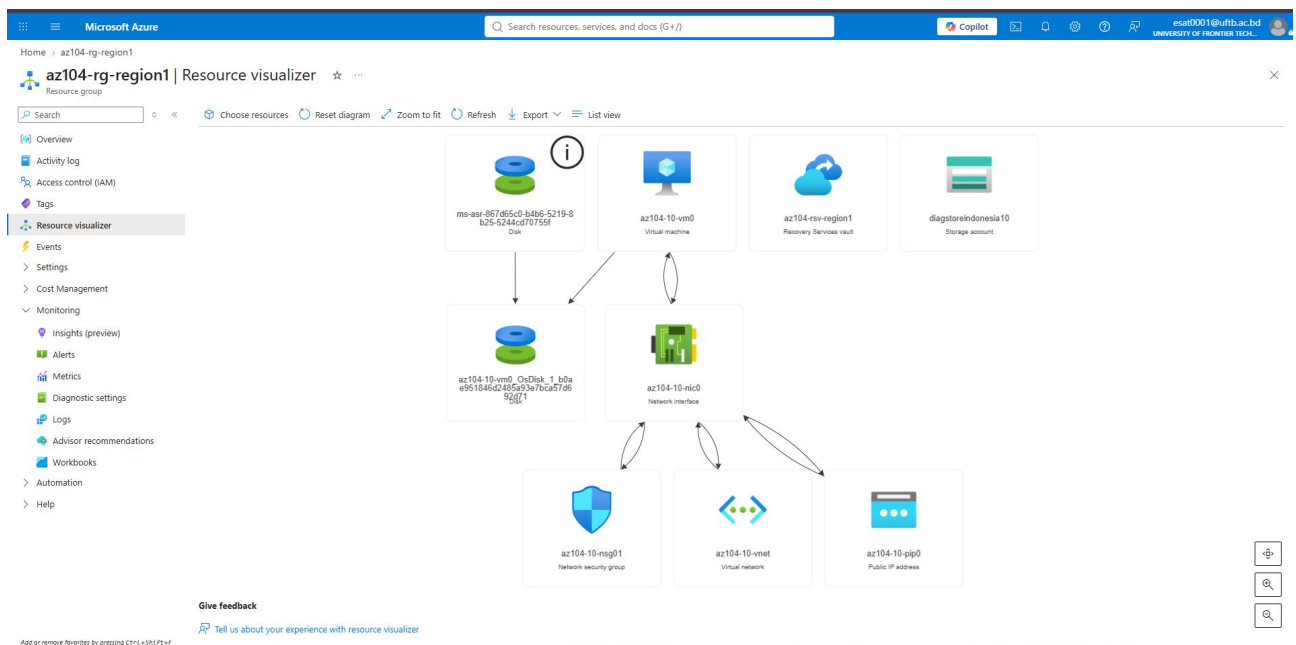
August 23, 2026

1. Executive Summary

This report outlines the implementation of complete data protection and disaster recovery strategies for Azure Virtual Machines (IaaS). The deployment encompasses two primary pillars: Azure Backup for operational, point-in-time recovery and Azure Site Recovery (ASR) for business continuity across paired geographic regions.

2. Architecture Overview

- **Primary Region:** Indonesia Central (az104-rg-region1)
- **Secondary Region:** Korea Central (az104-rg-region1-asr / az104-rsv-region2)
- **Protected Workload:** Virtual Machine (az104-10-vm0)
- **Core Protection Services:** Recovery Services Vaults, Azure Backup Policy, Azure Storage Diagnostics, and Azure Site Recovery.



3. Step-by-Step Task Execution & Verification

Task 1: Deploy Provisioning Infrastructure

Deployed the core primary infrastructure using ARM templates, establishing az104-10-vm0 inside az104-rg-region1 along with associated networking and storage resources.

The screenshot displays the Microsoft Azure portal interface. At the top, the 'Resource Manager' section is active, showing a list of resource groups. Below this, the 'az104-rg-region1' resource group is selected, and the 'Overview' tab is active. The overview shows the subscription 'Azure for Students' and the location 'Indonesia Central'. A table lists the resources within the group, including network interfaces, security groups, IP addresses, virtual machines, disks, and storage accounts.

Name	Type	Location
az104-10-nic0	Network Interface	Indonesia Central
az104-10-nsg01	Network security group	Indonesia Central
az104-10-pip0	Public IP address	Indonesia Central
az104-10-vm0	Virtual machine	Indonesia Central
az104-10-vm0_OsDisk_1_b0ae951846d2485a93e7bca57d692d71	Disk	Indonesia Central
az104-10-vnet	Virtual network	Indonesia Central
az104-rsv-region1	Recovery Services vault	Indonesia Central
diagstoreindonesia10	Storage account	Indonesia Central
ms-asr-867d65c0-b4b6-5219-8b25-5244cd70755f	Disk	Indonesia Central

Showing 1 - 9 of 9. Display count: auto

[Give feedback](#)

Task 2: Create Recovery Services Vaults

Created two Recovery Services Vaults in separate geographic locations to segregate local operational backups from disaster recovery management.

- **az104-rsv-region1** — Indonesia Central (Primary Vault for Local Backups).
- **az104-rsv-region2** — Korea Central (Secondary Vault for Disaster Recovery).

Name	Type	Security level	Resource Group	Location	Subscription
az104-rsv-region1	Recovery Services vault	Good	az104-rg-region1	Indonesia Central	Azure for Students
az104-rsv-region2	Recovery Services vault	Good	az104-rsv-region2	Korea Central	Azure for Students

Task 3: Configure Azure Virtual Machine Level Backup

Configured daily VM backup policies and bound az104-10-vm0 to az104-rsv-region1. Initiated an on-demand manual backup operation to validate backup pipeline connectivity.

Configuration Details:

- **Policy Name:** az104-backup-policy
- **Frequency:** Daily at 12:00 AM UTC
- **Retention:** Instant recovery snapshot (2 days), Daily retention (180 days).

Name	Subscription	Resource Group	Backup Pre-Check	Last Backup Status	Source scan integration	Source scan summary	Latest restore point	Policy name	Policy sub type	Details
az104-10-vm0	Azure for Students	az104-rg-region1	Passed	Success	Not configured	-	8/22/2026, 7:15:49 PM	az104-backup-policy	Standard	View details

Change Backup Policy

To change policy, please select a policy other than the one currently being used.

If recovery points are deleted as a result of changes to this backup policy, it may take up to 24 hours for them to show up as soft deleted. Similarly, if any soft deleted recovery points are undeleted as part of modifications to the backup policy, it may take up to 24 hours for the changes to reflect.

Modifying policy impacts existing recovery points as well and might get deleted. Deletion of recovery points in vault-archive might incur cost. [Learn More](#)

Changing from standard policy to enhanced policy can result in additional costs. Once changed to enhanced policy you cannot change again to standard policy type. [Learn More](#)

Policy subtype *

Backup policy *

Policy details

Full backup

Backup frequency
Daily at 12:00 AM UTC

Instant restore
Retain instant recovery snapshot(s) for 2 day(s)

Retention of daily backup point
Retain backup taken every day at 12:00 AM for 180 Day(s)

Consistency type

[Change](#) [Cancel](#) [Give feedback](#)

Task 4: Implement Diagnostic Settings

To support auditing and log retention, a diagnostic Storage Account (diagstoreindonesia10) was created. Vault logs and metrics were configured to archive to this storage destination.

Log Categories Forwarded:

- Azure Backup Reporting Data
- Addon Azure Backup Job Data
- Addon Azure Backup Alert Data
- Azure Site Recovery Jobs
- Azure Site Recovery Events

Home > Resource Manager | Resource groups > az104-rg-region1 > az104-rsv-region1

az104-rsv-region1 | Diagnostic settings

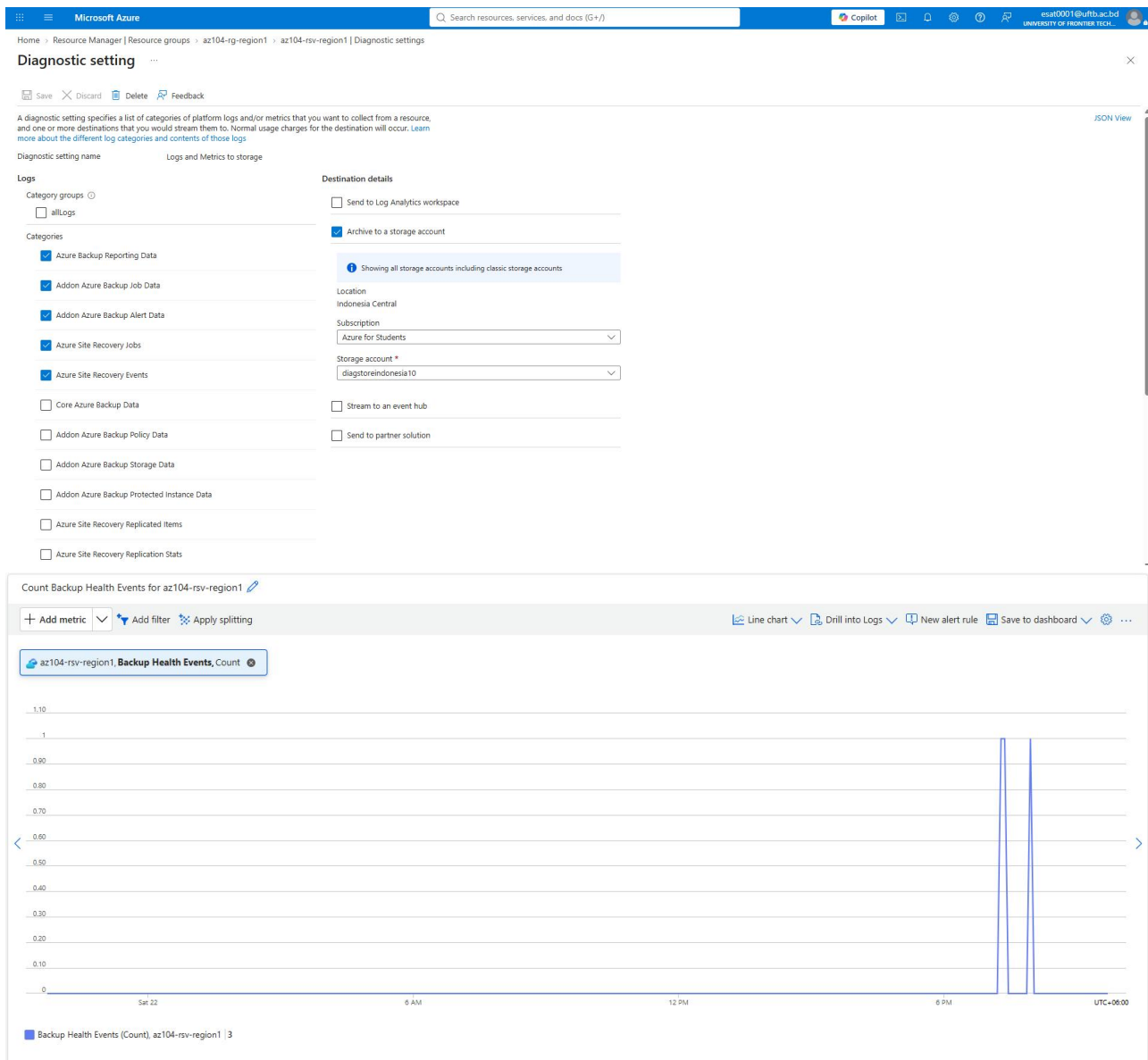
Diagnostic settings are used to configure streaming export of platform logs and metrics for a resource to the destination of your choice. You may create up to five different diagnostic settings to send different logs and metrics to independent destinations. [Learn more about diagnostic settings](#)

Name	Storage account	Event hub	Log Analytics workspace	Partner solution	Edit setting
Logs and Metrics to storage	diagstoreindonesia10	-	-	-	Edit setting

[+ Add diagnostic setting](#)

Click 'Add Diagnostic setting' above to configure the collection of the following data:

- Azure Backup Reporting Data
- Core Azure Backup Data
- Addon Azure Backup Job Data
- Addon Azure Backup Alert Data
- Addon Azure Backup Policy Data
- Addon Azure Backup Storage Data
- Addon Azure Backup Protected Instance Data
- Azure Site Recovery Jobs
- Azure Site Recovery Events
- Azure Site Recovery Replicated Items
- Azure Site Recovery Replication Stats
- Azure Site Recovery Recovery Points
- Azure Site Recovery Replication Data Upload Rate
- Azure Site Recovery Protected Disk Data Churn
- Azure Site Recovery Replicated Items Details
- Azure Backup Operations
- AllMetrics



Task 5: Implement Azure Site Recovery (ASR) Disaster Recovery

Configured cross-region disaster recovery for az104-10-vm0 from Indonesia Central to Korea Central. To bypass automation account creation restrictions under the Azure for Students subscription, update settings were explicitly set to Manage manually.

Target Configuration Summary:

- **Source Region:** Indonesia Central
- **Target Region:** Korea Central
- **Target Resource Group:** az104-rg-region1-asr
- **Target Virtual Network:** az104-10-vnet-asr
- **Extension Settings:** Manage manually

az104-rsv-region2 Resource group

Help me generate Terraform for this resource group configuration. Export resource groups using Bicep or Terraform +1

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Group by none

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
 - Insights (preview)
 - Alerts
 - Metrics
 - Diagnostic settings
 - Logs
 - Advisor recommendations
 - Workbooks
- Automation
- Help

Essentials [JSON View](#)

Subscription (move) : [Azure for Students](#) Deployments : 8 Failed, 13 Succeeded

Subscription ID : 73a6a025-34f7-4804-a613-553726e20799 Location : Korea Central

Tags (edit) : [Add tags](#)

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
az104-rsv-region2	Recovery Services vault	Korea Central
zvo2r1az104rsvasrcache	Storage account	Indonesia Central

Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Recovery Services vaults > az104-rsv-region2 | Replicated items

Enable replication

Source Virtual machines Replication settings Manage **Review**

Summary

Region	Indonesia Central
Virtual machine deployment model	ResourceManagement
Subscription	Azure for Students
Resource group	az104-rs-region1
Availability zones	-

Virtual machine selection summary

Virtual machines to replicate 1

VM settings

Region	Korea Central
Target subscription	Azure for Students
Target resource group	az104-rs-region1-asr
Post-failover virtual network	az104-10-vnet-asr
Replication policy	24-hour-retention-policy
Extension settings	Manage manually

Previous **Enable replication** Give Feedback

Operation	Status	Time	Subscription	User
Backup Protected Item	Succeeded	3 hours ago	Sat Aug 22 ...	Azure for Students
Backup Protected Item	Succeeded	3 hours ago	Sat Aug 22 ...	Azure for Students
Create Backup Protected Item	Succeeded	3 hours ago	Sat Aug 22 ...	Azure for Students
Create Backup Protected Item	Succeeded	3 hours ago	Sat Aug 22 ...	Azure for Students

4. Final Outputs & Verification

Output 1: ASR Replication Status (Secondary Vault)

Verification confirms that cross-region disaster recovery replication is active inside the secondary Recovery Services Vault.

Microsoft Azure

Home > Resource Manager | Resource groups > az104-rsv-region2

az104-rsv-region2 | Replicated items

Search resources, services, and docs (G+/I)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Getting started

Backup

Site Recovery

Protected items

Backup items

Replicated items

Manage

Backup policies

Backup infrastructure

Site Recovery infrastructure

Recovery Plans (Site Recovery)

Backup Reports

Monitoring

Alerts

Metrics

Feedback

Last refreshed at: 8/22/2026, 9:51:06 PM

Finished loading data from service.

Filter items...

Name	Replication Health	Status	Active location	Fallover Health
az104-10-vm0	Healthy	Protected	Indonesia Central	Warning

Showing 1 - 1 of 1 results.

Output 2: Virtual Machine Backup Status (Primary Vault)

Verification confirms that operational backup protection is actively maintained within the primary Recovery Services Vault.

Microsoft Azure

Home > Resource Manager | Resource groups > az104-rg-region1 > az104-rsv-region1 | Backup items

Backup Items (Azure Virtual Machine)

Search resources, services, and docs (G+/I)

Refresh

Add

Filter

Change policy

Bulk restore (Preview)

Export to CSV

Feedback

Show me the virtual machines that can be protected in this vault

You can now back up VMs across subscriptions with ease. Centralize protection and simplify multisubscription governance. [Learn more.](#)

Accelerate and simplify large-scale recovery by restoring multiple VMs with Bulk Restore (Public Preview). [Learn more.](#)

Filter items...

Name	Subscription	Resource Group	Backup Pre-Check	Last Backup Status	Source scan integration	Source scan summary	Latest restore point	Policy name	Policy sub type	Details
az104-10-vm0	Azure for Students	az104-rg-region1	Passed	Success	Not configured	-	8/22/2026, 7:15:49 PM	az104-backup-policy	Standard	View details

Previous Page 1 of 1 Next

Microsoft Azure

Home > Resource Manager | Resource groups > az104-rg-region1 > az104-rsv-region1 | Backup items > Backup Items (Azure Virtual Machine)

az104-10-vm0

Backup item

Backup now

Restore VM

File Recovery

Stop backup

Resume backup

Delete backup data

Restore to Secondary Region

Undelete

Feedback

Essentials

Recovery services vault : [az104-rsv-region1](#)

Subscription (move) : [Azure for Students](#)

Subscription ID : 73a6a025-347f-4804-a613-553726e20799

VM Subscription : [Azure for Students](#)

VM Subscription ID : 73a6a025-347f-4804-a613-553726e20799

Alerts (in last 24 hours) : [View alerts](#)

Jobs (in last 24 hours) : [View jobs](#)

Backup Pre-Check : Passed

Last backup status : Success 8/22/2026, 7:21:24 PM

Backup policy : [az104-backup-policy \(Standard\)](#)

Oldest restore point : 8/22/2026, 7:15:49 PM (3 hours) ago

Included disk(s) : [All disks](#)

Source scan integration : Not configured

Source scan summary : -

Recovery points

This list is filtered for last 30 days of recovery points. To recover from recovery point older than 30 days, as well as vault-archive, [click here](#).

Long term recovery points can be moved to vault-archive. To move all 'recommended recovery points' to vault-archive tier, [click here](#).

CRASH CONSISTENT 0

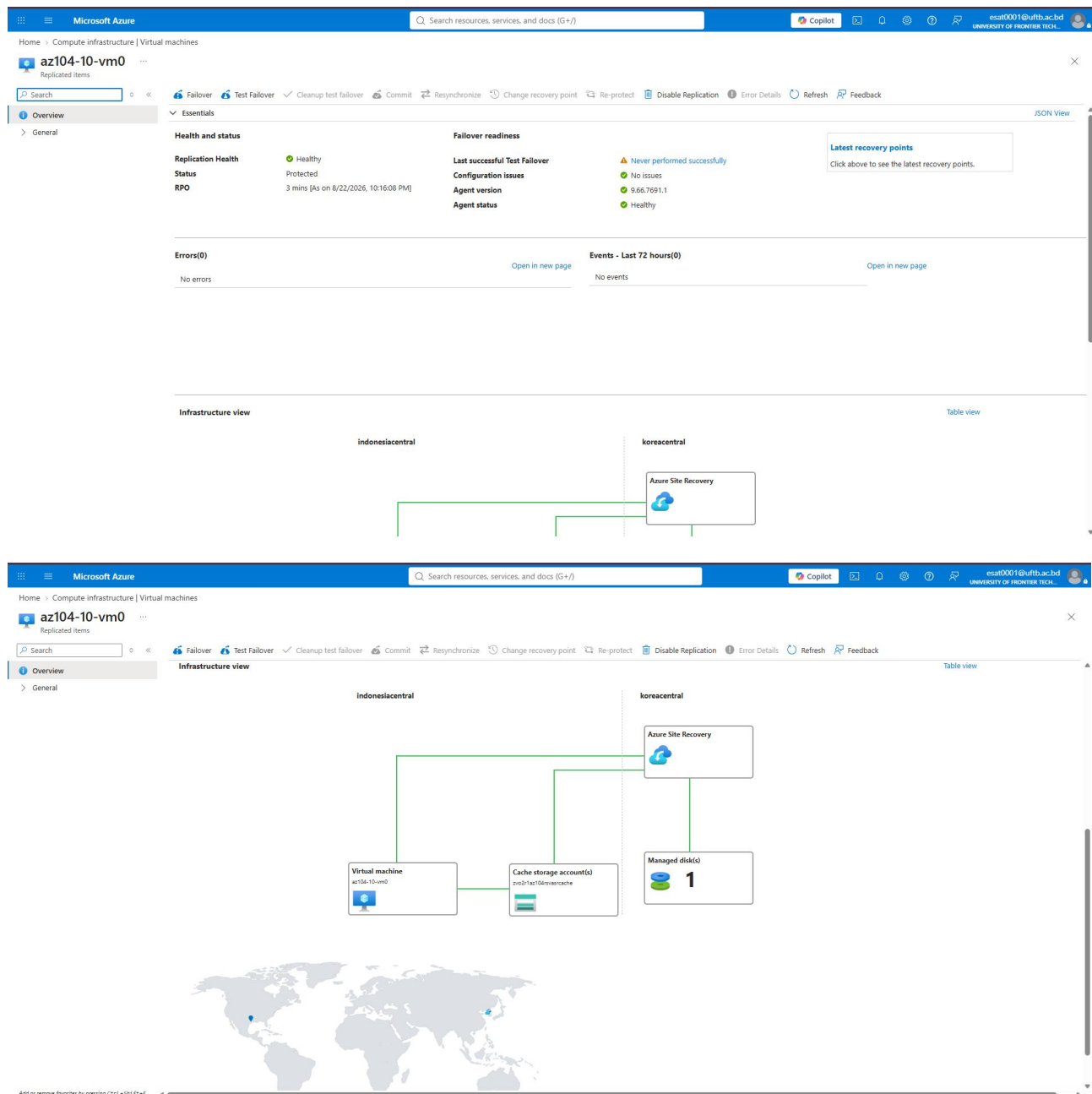
APPLICATION CONSISTENT 1

FILE-SYSTEM CONSISTENT 0

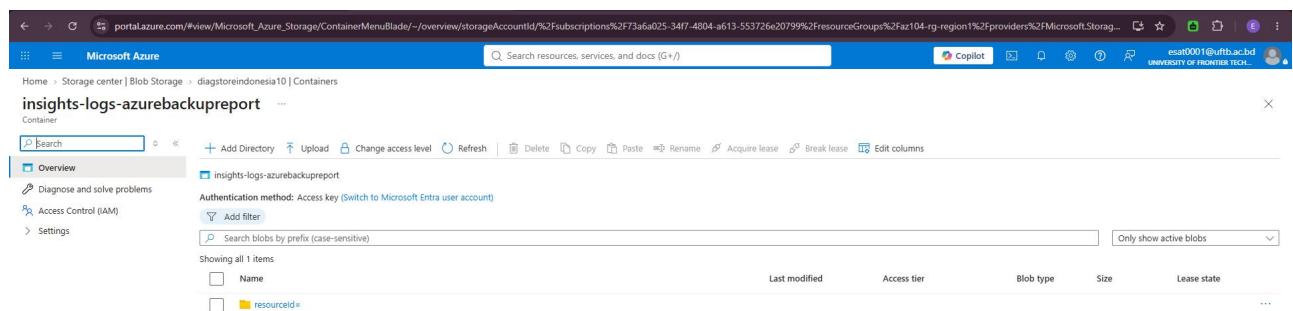
Creation time	Consistency	Recovery type	Recent scan status
8/22/2026, 7:15:49 PM	Application Consistent	Snapshot and Vault-Standard	Not available

Output 3: Cross-Region Topology Map

Visualization of the active protection link established between the primary and target regions.



The presence of the insights-logs-azurebackupreport container and its inner telemetry directories validates that vault diagnostics are actively writing logs to diagstoreindonesia10.



Demonstrates that vault diagnostic logs and metrics are being successfully exported and segregated into dedicated storage containers for compliance, auditing, and long-term security monitoring.

diagstoreindonesia10 | Containers

Storage account

Search containers by prefix

Only show active containers

	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	8/22/2026, 7:29:11 PM	Private	Available	-
☐	insights-logs-addonazurebackupjobs	8/22/2026, 8:02:11 PM	Private	Available	-
☐	insights-logs-azurebackupreport	8/22/2026, 8:02:11 PM	Private	Available	-

Till Now total cost.....

Microsoft Azure

Search resources, services, and docs (G+/I)

esat0001@ufhb.ac.bd

Home

MST. AKHI

Billing account

Overview

Access control (IAM)

Billing scopes

Diagnose and solve problems

Cost management

Billing

Invoices

Payment methods

Payment history

Reservation transactions

Billing profiles

Benefits

Products + services

Settings

Support + troubleshooting

Amount due

\$0.00

No payment needed

View invoices

Upcoming invoices

Available on 8/9/2026

Billing period 8/1/2026 - 8/31/2026

Pre-tax total so far \$15.49

Invoices over time

No invoices in last 6 months

Total amount \$0.00

Spending rate and forecast

Current charges \$15.49

Forecast \$24.28

View details

This month's top products by charges

Premium SSD Managed Disks \$8.45

IP Address - Standard IPv4 \$3.97

Virtual Machines Dv3 Series \$1.91

Other purchases \$1.15

View details

Billing alerts (0)

No billing alerts to display

Shortcuts

Download usage and prices

Manage payment methods

View billing subscriptions

Share monthly invoice by email

View your support plan

Give feedback

Credits remaining

\$84.51

\$15.49 used of \$100

Used Amount remaining

View credits

5. Conclusion

The lab successfully demonstrated end-to-end data protection for Azure IaaS Virtual Machines. Operational recovery was achieved through automated backup policies, auditability was established via diagnostic storage archiving, and regional fault tolerance was successfully implemented using Azure Site Recovery.